Lab 1 – Smart City Product Description

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1 Introduction

People often must pack up their lives, in search of better opportunities, and travel across the world to an unfamiliar location. As a result, cities all over the world have begun to see an increase in population size. According to number analytics, by 2050 the UN predicts that 6.5 billion people will live in urban areas with most of these people moving to developing countries which intensifies the demand for public services (number analytics). This can lead to overcrowding in the communities that have been established in these cities for a long time. As seen in Figure 1, the percentage of the population that has been living in urban areas has increased from the year 1990 to 2014 and it is predicted to keep rising. We see that

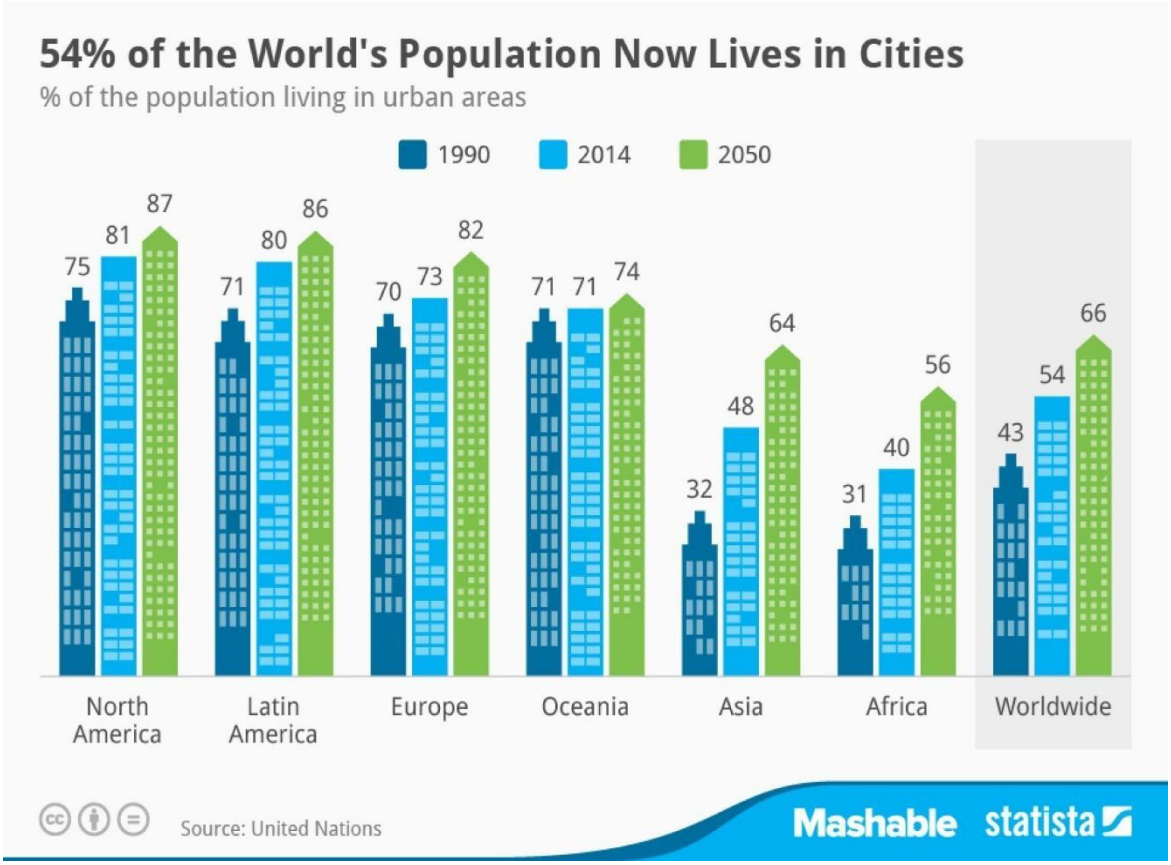

Figure 1: Population growth through 2050

more people are moving to urban areas, which often have limited space. This is a trend that is happening at a global level which makes it important that it is addressed as overcrowding and rapid urbanization can lead to serious problems. Overcrowding tends to force people to live in proximity to one another which can exacerbate ongoing issues such as pollution and unsafe living conditions. This can cause them to develop health issues such as respiratory illnesses and mental health problems which then leads to higher mortality rates in these areas. The residents are affected by traffic congestion, poor living conditions, and increased crime which all make their lives harder. All these issues negatively affect the human mind, making it extremely difficult to simply live in peace. Due to the impact of the problem, it is critical that city residents have access to tools and services to combat the rapid urbanization currently ongoing in their communities.

One main obstacle standing in the way of fixing the problem is the current condition of the infrastructure in place. Current infrastructure is not sustainable to deal with rapid urbanization and there is little investment towards those systems. This means that services that help residents can be underfunded and unavailable to support the community. There is also minimal coordination between local government agencies. When groups are not communicating and working effectively it hinders the ability of the government to support the residents. Lastly, every initiative to address the issues at hand has always relied on governments and city officials. This leaves no voice for the people to speak about issues unless they can access a city hall meeting. If there was a centralized platform to share information, where everyone in the city could share different types of data to make their city operate smoothly, then it would be easier to manage and address the scale of issues communities are facing.

By providing a platform for communities where people from all walks of life can come to interact, talk about their surroundings and share information we aim to create cleaner, secure and intelligent cities. Smart City will provide residents with the application needed to update their communities. By leveraging data gathered from multiple sources such as forums, navigation and reports, Smart City will supply residents and city planners with all the information needed to facilitate the proper cleaning and security of their neighborhoods, to create smarter cities. With Smart City our goal is to leverage this information to combat the problem of overcrowding and the issues that arise from overcrowding in cities.

2 Product Description

Smart City is an app that connects people with the city around them, offering them navigation tools, forums and insights in an efficient way. Residents can use Smart City to move around in their city or any unfamiliar environment. City Planners can use our application to gather information for smart urban development which the residents benefit from and contribute to. It aims to break the barrier to data collection by implementing crowdsourcing methods and letting the residents have a voice in their community.

2.1. Key Product Features and Capabilities

Key product features include an engaging user interface with a community forum, navigation system, insights and efficient data handling by processing and storing collected data. The community forum will allow users to interact with one another and city officials, with the focus on encouraging collaborative effort between community members. This helps foster an environment where people want to help in their communities and create smarter cities.

Smart City is significant and unique as we aim to eliminate the barriers to data collection while providing a one-stop solution that enables citizens to navigate their communities in a simpler manner. With a user-friendly and engaging navigation system, Smart City will aim to incentivize traffic coordination and reports between citizens to aid in congestion, road hazards and obstructions of travel. This works with our data collection which Smart City will record and can be used to analyze for aid in city planning. With real time alerts and insights, Smart City will keep residents in the loop on what is going on in their community which they can use to plan accordingly.

Smart City looks to solve the problem of rapid urbanization and overcrowding that has been trending globally. This is a critical problem to address as it has negative health effects on the populations of urban cities. They suffer from traffic jams, poor quality housing, inadequate waste management, water scarcity and increased crime rate. By implementing a software-based solution, we aim to provide citizens with the tools they need to navigate their community without any hassle.

2.2. Major Components (Hardware/Software)

Hardware needed to support the solution includes a smartphone (iOS or Android) for users to access the application for navigation and community hub and Internet connectivity is needed to engage with the application. Users will use their smartphone to sign-in to the app and engage with the features provided. Internet is required to access the community forum to make posts and interact with other users. In Figure 2 the Major Functional Components Diagram is shown. This illustrates how the application is intended to operate along with features that will be implemented.

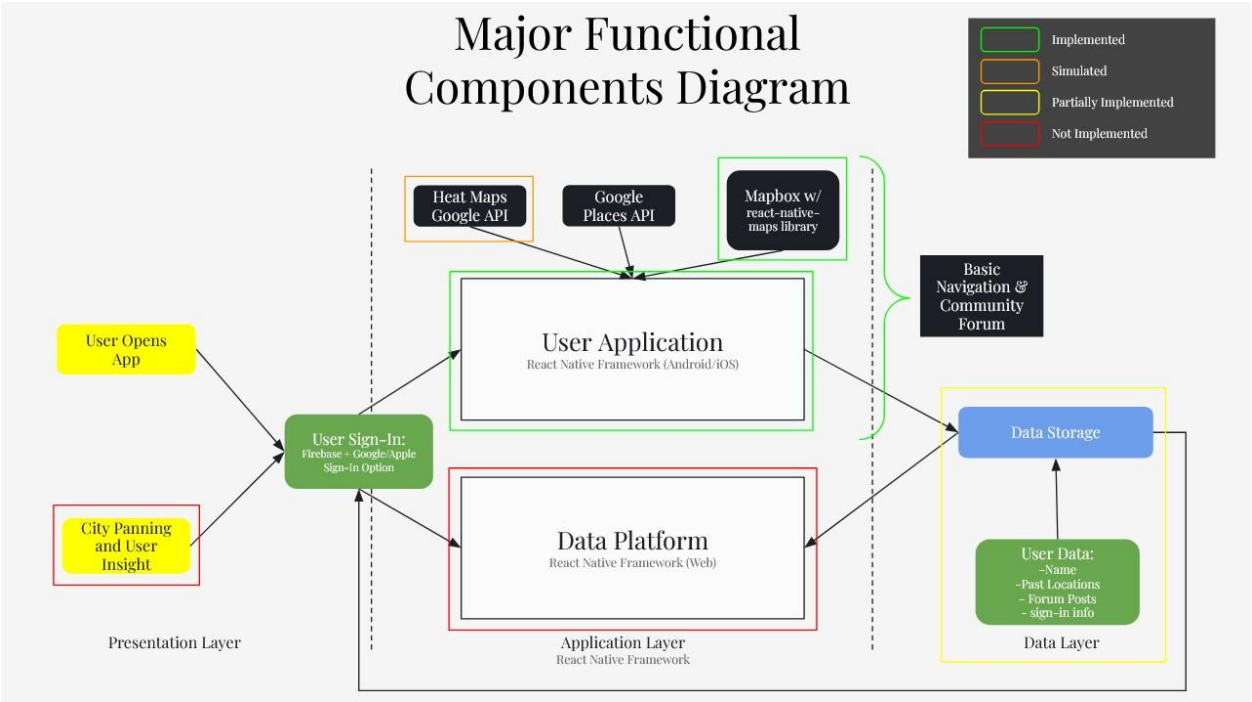

Figure 2: Original Major Functional Components Diagram

The Major Functional Components Diagram is split into three different sections. We first have the presentation layer which allows users and city officials to engage with the application. Then we have the application layer which provides the logical components that make up the application. This includes the APIs that will be interacting with the user application and handling navigation. Lastly, we have the data layer which houses the database we will use for data storage.

Smart City will be using a couple of different pieces of software to implement. First, the user application and data platform modules will be developed using Expo, which is a React Native Framework, alongside using Bun for its runtime toolchain which includes a bundler and test-runner. The user application will interact with APIs such as Mapbox and OpenStreetMap to aid in building out the navigation system. We will be using JavaScript for front end and

Node.js/Bun for backend development along. We will use MySQL Workbench to implement our database while hosting on a virtual machine.

3 Identification of Case Study

3.1. Who It's For

There are many groups of people that can benefit from Smart City. The first group of people are the local city residents and commuters who simply want real time traffic, event and safety notifications straight to their device. The second group of people are community members who want to share or check local updates quickly. In future implementations, our final group of people would be the city officials who would be using our data platform to garner insights and improve decision making on long-term city development.

3.2. Why

Smart City brings many features and improvements that will make their lives easier to navigate day to day. With real time traffic and event analysis, residents get easier commutes and lower amounts of traffic issues arise. By accessing the Smart City community forum members feel empowered to stay informed about what is happening around them, which incentivizes them to use Smart City. By combining navigation system data with crowdsourced community input we aim to compile data that can aid in future city developments. In future implementations, our goal is to add machine learning methods to analyze data gathered for city officials to use.

3.3. Case Study Group

The case study group can be a small group of ODU students (about 5 – 10 students) along with nearby residents from the surrounding neighborhoods like Colonial and Park Place. The group of students along with the residents will play their roles respectively as the main user base

of the application. There can also be ODU faculty involved in the case study and they can take the role of the "City Planner" in our case. They can use the prototype app on their mobile devices to maneuver around their surroundings. We can take their data and experience as feedback to make improvements on the prototype.

3.4. Potential Future Uses

For potential future uses, we can expand the user base outside of ODU campus and its neighboring regions to include more people. First, we can scale Smart City and make it available for the City of Norfolk to use. Once the application has been set up and it is operating well in that area it can be expanded to serve the greater Hampton Roads area, encompassing Virginia Beach, Chesapeake and Hampton. This would allow us to potentially tap into the local business around the area to serve them with the tools they need to come together and build community. They could join Smart City and share their own events that they got going on to boost the community's engagement with their business. Future uses would also include allowing city planners to access the data collected and having tourists who are visiting new locations using our app for navigation and local updates.

4 Smart City Product Prototype Description

Our priority when designing Smart City was to provide city residents and locals with tools and resources to navigate through their community and connect with one another efficiently and safely. The prototype for Smart City will not have every feature that we had intended for but will still be able to demonstrate how it can work as a solution to the stated problems discussed in Section 1 of the document.

4.1. Prototype Architecture (Hardware/Software)

The prototype for Smart City will include the following major components which are shown in Figure 3. Some of the key changes we had to make for the prototype were combining the user login for both residents and city planners to access it, narrowing down which APIs we were going to use and must implement for use in the navigation system and lastly getting rid of the long-term data platform web application. These decisions were made to streamline production of the prototype to ensure a working product within the given time frame. Since the removal of the web application all users will need to access Smart City is a smartphone with connection to the Internet and iOS/Android app markets. One new incorporation is the use of ElysiaJS as the framework for building our backend server application. Elysia works well with Bun, which is another runtime toolkit we are using for assembling the project. Bun can work as a package and build manager along with offering a testing suite that works with Jest enabling us with all the tools needed to build our application.

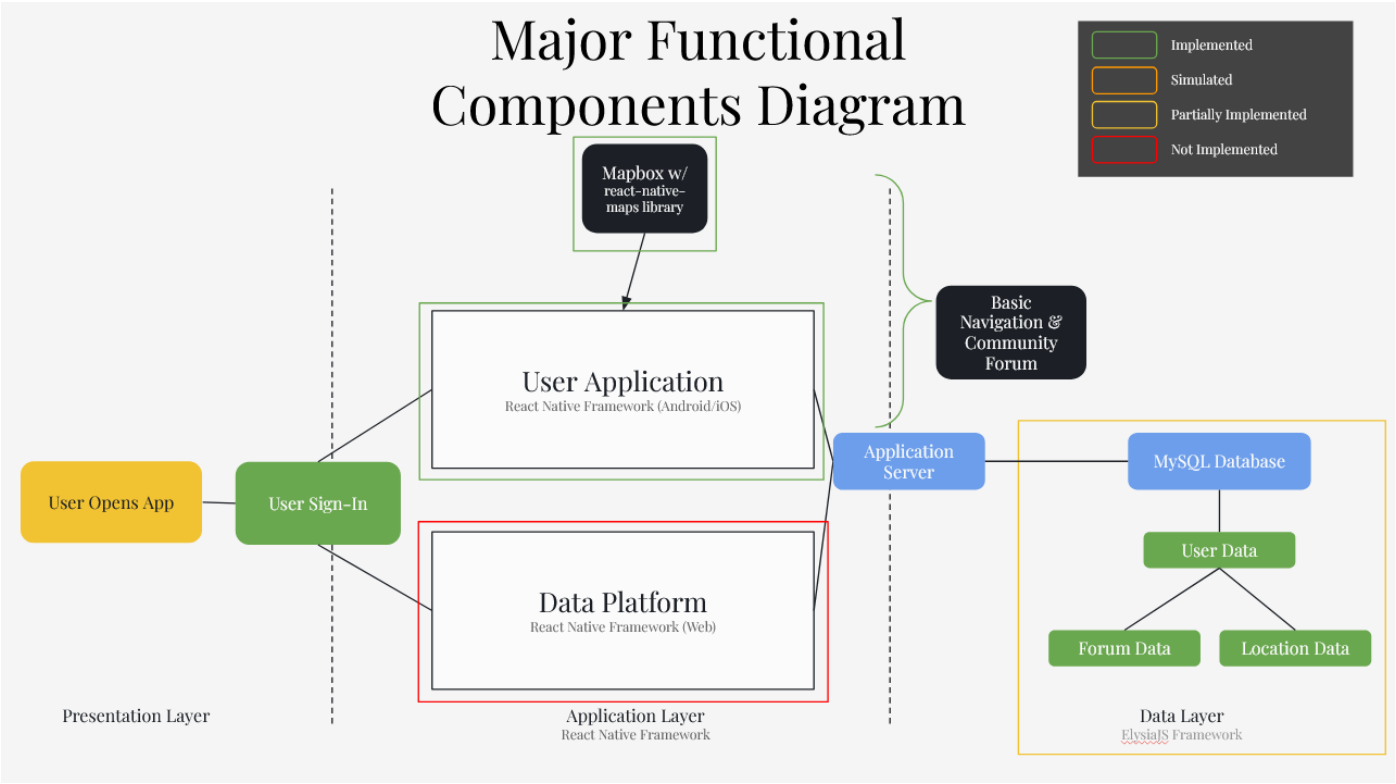

Figure 3: Prototype Major Functional Components Diagram

4.2. Prototype Features and Capabilities

We had to go through and revisit some of the features we had listed to deliver a working product within the given timeframe. Most of the features from the real-world product will be either fully implemented or partially implemented with some tweaks done to ensure completion. A handful of features unfortunately will not be implemented in the prototype. The adjustments to the features as we progressed from the real-world product to the prototype are showcased in Table 1. Some of the main features that had to be altered were the Navigation system and the web data platform. The navigation system will still work as intended but we had to drop the implementation of the dynamic queue as it would be too complex for the scope of the project. Another feature that had to get revamped was the data platform as we had to drop the long-term web application portion of the system. With the time that we are allotted we simply could not commit to focusing on this feature in the manner that we had hoped for. We will still be implementing the database for storing our data.

Table 1: Smart City RWP v. Prototype Features

Feature Table: RWP vs. Prototype

Category	Features	RWP	Current Prototype	Reasoning
Account Management	Login/Authentication	Fully Implemented	Fully Implemented	Essential for prototype
	Account Creation/Deletion	Fully Implemented	Fully Implemented	Essential for prototype
	Manage Account Access/Permissions	Fully Implemented	Fully Implemented	Essential for prototype
User Application	Access Community Forum	Fully Implemented	Fully Implemented	Essential for prototype
	Access Dynamic Queue System for Navigation	Fully Implemented	Partially Implemented	We removed the Dynamic Queue implementation as it would be too complex for the given time frame. Still implementing the basic navigation.
	Alerts and Notifications	Fully Implemented	Partially Implemented	Simulated feature until we figure out how to handle API calls.
	Add Parking locations	Fully Implemented	Not Implemented	No time given scope of project. Could reconsider

Feature Table: RWP vs. Prototype Continued

Category	Features	RWP	Current Prototype	Reasoning
Data Platform	Central Information Hub	Fully Implemented	Partially Implemented	Still will implement the database for storing information in. Will drop long-term data platform due to time constraints.
	Data Analytics and Insights	Fully Implemented	Not Implemented	Dropped due to no longer implementing the long-term data platform
	Visualize Data	Fully Implemented	Not Implemented	Dropped due to no longer implementing the long-term data platform
	Data Anonymity	Fully Implemented	Fully Implemented	Essential for prototype

Our alerts and notifications will be partially implemented as a simulated feature until we can resolve how to handle API routes for real-time consistent alerts. Another feature that will not be implemented is the add parking lot feature which would give businesses the ability to add that to their home page for customers to have. This was not one of the core features we wanted to focus on so we decided to drop it for now and later down the line we could reconsider if time permits.

Despite these changes to the real-world product, the features that we will include in the prototype for Smart City still allow residents to navigate and coordinate in their city with ease. It will provide security with user authentication and data anonymity to ensure privacy. Users will be able to receive alerts and notifications regarding their accounts and for features such as maps and forums.

4.3. Prototype Development Challenges

As stated earlier in Section 4.2, some of the features from the real-world product version of Smart City will not be implemented in the prototype. The algorithm for the dynamic queue would have been too complex to add and does not affect how the navigation system works with

the maps if it is not implemented. Other challenges that we may face when developing the application are learning new technologies and following the sprint plan. Many of us are learning new tools such as Bun and MySQL for the first time. Trying to learn these tools as we begin to build out the project can lead to technical issues that can slow the team down as we work to find a solution. This ties into the next challenge of following the sprint plan. When we hit inevitable roadblocks in development it will be crucial to manage our time wisely as we move from sprint to sprint. Due to the time constraints not all the features will be implemented and prioritizing the right tasks becomes crucial. By deciding on a minimum viable product to showcase, we can curb some of these issues by making sure everyone agrees on what features are being built.

5 Glossary

Internet of Things (IoT): systems that combine sensing, computing, communication, and actuation to connect physical/cyber objects

Application Programming Interface (API): a set of features and rules that lets software interact with other software (a "contract" between programs)

Crowdsourced Reporting: gathering information by soliciting contributions from a large group of people, typically via the internet

Privacy-by-Design ("Privacy-First Design"): building systems so that, by design and by default, only the personal data necessary for a purpose are processed

Open Data (Open Data Access): data anyone can freely use, re-use, and redistribute, with at most attribution/share-alike requirements

Predictive AI / AI-Powered Forecasting: using data, statistical algorithms, and ML to estimate the likelihood of future outcomes

Anomaly Detection: identifying data patterns that deviate from expected behavior

Access Control: controls that enforce authorized access to information and system resources in line with policy

Citizen Feedback Loop: a two-way process where institutions act on and report back about citizen input to sustain engagement and trust.

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